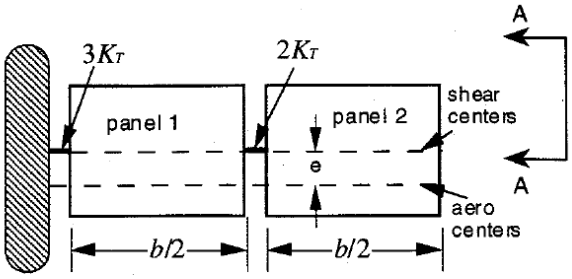
# MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
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| DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGNEERING | | | |
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| ~~MID-TERM EXAMINATION~~/ END SEMESTER EXAMINATION/ ~~MAKE-UP EXAMINATION~~ # | | NOVEMBER, 2024 | |
| **Level** | **B.Tech/~~M.Tech./MCA/M.Sc~~. #** | **Curriculum** | **2018 / 2022 #** |
| **Program Name** | **B.Tech Aeronautical Engg.** | **Course Type** | **~~C/FC~~/PE/~~OE~~** |
| **Course Name** | **Aeroelasticity** | **Course Code** | **AAE 4044** |
| **Semester** | **VII Semester** | **Max Marks** | **50** |
| **Date** | **13.11.2024** | **Duration** | **3 Hrs** |

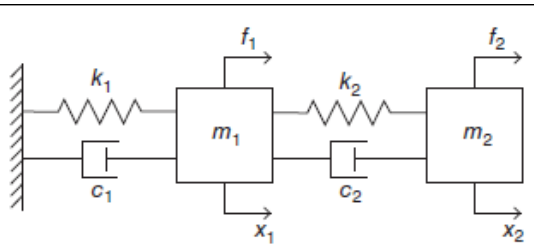
**Instructions:**

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

| Q. No. | Question | Marks | NBA | | IET | | BL |
|---|---|---|---|---|---|---|---|
| | | | CO | PO | CLO | AHEP4 LO | |
| 1A | Using an approximate deflection mode shape given below, $\theta = 2\left(\frac{y}{l}\right) - \left(\frac{y}{l}\right)^2$ compute the torsional divergence speed of a uniform wing by using Rayleigh-Ritz method and Strip theory, and compare with exact solution | **4** | 2 | 1, 4, 12 | **2** | **1, 2, 3** | **3** |
| 1B | Interpret, the impact on pitch angle and lift, if an aircraft is made to go faster and faster. (Note: It leads to an unstable flight condition in static aeroelasticity) | **3** | 3 | 1, 4 , 12 | **3** | **1, 2, 3** | **3** |
| 1C | Interpret the reason for swept back wings having very controlled divergence and more possibility of control effectiveness (reversal) as compared to swept forward wings. | **3** | 3 | 1, 4 , 12 | **3** | **1, 2, 3** | **3** |

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| **2A** | The two airfoil sections illustrated in figure 1 below are mounted on a rigid shaft, allowing for rotation but not vertical movement (plunge). Both sections are uncambered and start with the same angle of attack, denoted by $\alpha_0$. The rotation of each section relative to the other is restricted by a torsion spring, which simulates the wing structure's resistance to relative rotation between sections. Additionally, a torsion spring restricts the rotation of the inboard section relative to the wall. The figure displays the locations and values of these torsion spring constants, along with the planform geometry. Estimate the divergence dynamic pressure $q_D$ of the system.<br>  <br><br>Figure 1 | **5** | 3 | 1, 4, 12 | **3** | **1, 2, 3** | **4** |
| **2C** | It is observed that reversal speed is independent of the elastic axis offset *'e'*. Write down the appropriate reason for this? | **3** | 3 | 1, 4 , 12 | **3** | **1, 2, 3** | **2** |
| **2B** | Write the boundary conditions of a wing idealised as a cantilever beam under torsional deformation with uniform distributed load (with diagrammatical representation). | **2** | 2 | 1, 4 , 12 | **2** | **1, 2, 3** | **2** |
| **4A** | Control reversal is not an eigen value problem in the same sense as divergence. Indeed the characteristic dynamic pressure associated with the rigidly deflected flap is still dynamic divergence pressure. With the knowledge of the control reversal problem with ailerons, anticipate the same | **4** | 3 | 1, 4 , 12 | **3** | **1, 2, 3** | **5** |

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| | to other control surface, in this case the spoilers. Use mathematical interpretation for clarity. | | | | | | |
| **4B** | A uniform member of the cantilever form has the field equation given as:<br>$$\theta'' + \frac{\omega^2}{a^2}\theta = 0, \qquad a^2 = \frac{GJ}{I_1}$$<br>Calculate its natural mode shapes and frequencies using exact solution and boundary conditions | **3** | 1 | 1, 2, 3, 4, 12 | **1** | | **3** |
| **4C** | Explain the underlying principles involved in Galerkin's method and the Hamilton's Principal. | **3** | 2 | 1, 4, 12 | **2** | **1, 2, 3** | **3** |
| **3A** | A binary aeroelastic system (in SI units) takes the form<br>$$\begin{bmatrix}130 & 0\\ 0 & 10\end{bmatrix}\begin{bmatrix}\ddot{\theta}\\ \ddot{\gamma}\end{bmatrix} + \begin{bmatrix}6V & 0\\ -3V & V\end{bmatrix}\begin{bmatrix}\dot{\theta}\\ \dot{\gamma}\end{bmatrix} + \begin{bmatrix}k & 3V^2\\ 0 & 2k-3V^2\end{bmatrix}\begin{bmatrix}\theta\\ \gamma\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$<br>For the stiffness value *'k'* of 1.173 × 105 N/m, formulate the quartic equation using Routh-Hurwitz criterion. | **5** | 5 | 1, 3, 4, 9, 12 | **5** | **1, 2, 3** | **3** |
| **3B** | Continuing calculations from Q.10, estimate the critical flutter speed $V$ and the corresponding flutter frequency. | **2** | 5 | 1, 3, 4, 9, 12 | **5** | **1, 2, 3** | **4** |
| **3C** | Answer the following: (a) With some logical reasoning, explain, the possibility to Combat flutter with mass distribution (b) Compose the standard definition of divergence dynamic pressure. | **3** | 3 | 1, 4, 12 | **3** | **1, 2, 3** | **3** |
| **5A** | Consider the example of the classical 2 DOF dynamic system shown in figure 2 below with two masses $\boldsymbol{m_1}$ and $\boldsymbol{m_2}$. Using the standard Lagrange's equation, formulate the equation of motion of the system in matrix format. | **5** | 2 | 1, 4, 12 | **2** | **1, 2, 3** | **5** |

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| |  Figure 2 | | | | | | |
| **5B** | Show that the influence coefficient is symmetrical in both matrix and function format. | **3** | 1 | 1, 2, 3, 4 , 12 | **1** | **1, 2, 3** | **2** |
| **5C** | Write down the primary source of antisymmetric loadings (lift distribution)? | **2** | 4 | 1, 4 , 12 | **4** | **1, 2, 3** | **2** |

**MARK DISTRIBUTION (CO-wise & BTL-wise)**

| CO / CLO | Total Marks |
|---|---|
| **1** | **06** |
| **2** | **14** |
| **3** | **19** |
| **4** | **02** |
| **5** | **09** |

| BL | Total Marks |
|---|---|
| **2** | **03** |
| **3** | **23** |
| **4** | **02** |
| **5** | **00** |